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PERFORMANCE SPECIFICATION

GENERATOR, SIGNAL, VHF
SG-1170/U

This specification is approved for use by the USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines a VHF Signal Generator that operates over a frequency range of 400KHz to 512 MHz and is capable of producing various signal function waveforms in the VHF spectrum, hereafter referred to as the equipment.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Communications- Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A: FSC 6625

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2.2 Government Documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Performance characteristics.

3.1.1 Output frequency. The equipment, shall provide an amplitude modulated (AM), frequency modulated (FM), and pulse modulated (PM) output covering a selectable frequency range of 400 Khz to 512 MHz over a maximum of 6 frequency bands.

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3.1.1.1 Physical characteristics. The design and physical characteristics shall be such that the equipment can be easily installed in standard equipment racks 24 inches maximum height, 21 inches maximum depth and 22.5 inches maximum width. The maximum weight shall be 50 pounds. The equipment shall be readily accessible for maintenance and/or calibration.

3.1.2 Preconditioning. Each deliverable equipment shall be subjected to a minimum of 96 hours burn-in prior to Group A testing. During the last 24 hours of the burn-in period the equipment must operate failure free. Equipment that fails during the last 24 hours of burn-in shall be repaired and resubmitted for testing until it successfully passes a 24 hour period without failure.

3.1.3 Output frequency. The output frequency range of the equipment, shall extend from 400 Khz to 512 MHz, selectable in 1 Khz steps, over a maximum of six frequency bands.

3.1.3.1 Output frequency response. The minimum required accuracy shall be ±0.5 percent of the output frequency and the output level shall not vary more than ±2dB over any individual frequency band.

3.1.3.2 RF spurious output. The total harmonic content of the rf output shall be at least 40 dB below the level of the unmodulated carrier.

3.1.3.3 RF noise level. The rf noise level shall be at least 35 dB below the level of the unmodutated carrier.

3.1.4 Amplitude modulation (AM). The equipment shall have the capability of amplitude modulation of the carrier signal with an internal or external sine or square wave at a frequency of at least 20 Hz to 20 KHz.

3.1.4.1 External AM sensitivity. An external sine or square wave of less than 5 volts peak-to-peak shall produce a modulation level of at least 95 percent.

3.1.4.2 Percent modulation. The equipment shall be capable of 0 to 95 percent amplitude modulation by both an external audio tone and an internal audio tone.

3.1.5 Amplitude modulation, internal. The internal amplitude modulation frequency shall be selectable between the frequencies of 400 Hz and 1000 Hz with a frequency accuracy of ±5 percent.

3.1.5.1 Modulation distortion. The distortion of the amplitude modulated carrier shall not exceed 2 percent.

3.1.6 Pulse modulation (PM). The equipment shall be capable of being pulse modulated by an internal or external signal at repetition rates of 50 to 5000 pulses per second with pulse widths varying from 10 to 40 microseconds.

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3.1.7 Frequency modulation (FM) deviation. The equipment shall have both an internal and external capability of 0 to 40 KHz deviation.

3.1.7.1 FM internal spurious AM. The spurious amplitude modulation shall not exceed 5 percent, when the signal generator is frequency modulated at any frequency with deviation set at 40 KHz.

3.1.7.2 Carrier frequency shift. Deviation of the frequency spectrum up to and including 25 KHz with modulating frequencies cited in this specification shall not cause a carrier shift in excess of 0.005 percent of the carrier frequency.

3.1.7.3 Modulation voltage. The voltage required to produce a 10 KHz deviation in the range from 100 Hz to 15 KHz shall not differ more than ± 10 percent from the voltage required at 1000Hz and shall not exceed ± 20 percent from 15 KHz to 30 KHz.

3.1.8 RF voltage output. As a minimum the output voltage range shall be variable from 0.1 microvolt to 1.0 volt rms across a 50 ohm load, and the accuracy shall be within ± 1dB of the indicated value.

3.1.8.1 Standing wave ratio (SWR). The SWR of the rf output level shall be less than 1.2.

3.1.9 Output attenuation. The output attenuation range shall be at least l30 dB, with an additional l0 dB controlled by a vernier that is calibrated in l dB steps. The vernier shall have an accuracy of ± 0.2dB or better.

3.1.10 Output impedance. The output impedance of the equipment shall be 50 ohms.

3.1.11 Input impedance. The input impedance of the equipment shall be 600 ohms.

3.2 Qualification. The equipment shall be constructed from parts and materials that shall provide high reliability under varying operational environments, ease of operation, and ruggedness. Radioactive materials (e.g. luminous dial/markings, lenses, etc.)shall not be used in the manufacture/construction of the equipment (see 6.2).

3.2.1 Safety. Unless otherwise specified, the equipment shall be designed and built to best commercial practices for electronic test equipment. Operating and maintenance personnel shall be protected from mechanical and electrical hazards by insuring that the equipment is properly shielded, marked, grounded and designed to prevent accidental contact with moving mechanical parts, such as fans and gears.

3.2.1.1 Shock protection. The equipment shall provide protection from HF voltages or currents that are capable of causing burns. Shock protection shall be provided if the open circuit voltage between any accessible part of the test equipment and ground, or any other accessible

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part, exceeds 30V rms, 60V dc, or 24.8V dc if it is interrupted at a rate of 10Hz to 200Hz. Parts of equipment which are part of an operators normal use or could become accessible upon removing a cover, opening a door, adjusting a control, replacing a fuse, etc., shall not be capable of rendering an electrical shock.

3.2.1.2 Open circuit protection. The open circuit potential between an accessible part and either ground or another accessible part shall not exceed the limits of paragraph 3.2.1.1. To verify this requirement, an open circuit voltage test shall be performed in accordance with paragraph 4.5.1.

3.2.1.3 Leakage current protection. If the open circuit voltage potential between two accessible points exceeds the limits of 3.2.1.1, the measured current shall not exceed 3.5ma when measured in accordance with 4.5.2

3.2.1.4 Grounding. The design and construction of the equipment shall ensure that all exposed metal parts such as enclosures, guards, controls, switches, etc. are at earth ground potential. Provisions shall be made to externally connect the equipment to a ground connection that has a low impedance path to earth ground.

3.2.1.4.1 Radio frequency interference (RFI). If RFI filters are used, they shall contain only components such as capacitors and inductors between each side of the power line and a safety ground which will ensure that no more than 3.5ma (rms) of current will flow to the chassis under the most adverse conditions.

3.2.1.4.2 Power cables. Cable grounding for any external interconnecting cable shall include a ground wire which can be terminated at both ends of its connection. Except for coaxial cables, the grounding shield shall not be used for grounding purposes.

3.3 Mechanical design. The detailed mechanical and electrical design of the equipment shall be accomplished by the contractor subject to the requirements of this specification. Unless otherwise specified, the equipment shall be modified commercial-off-the-shelf (COTS) equipment (Type II) that conforms to one or more military requirements not available in Type III. The equipment shall be capable for use as bench top, rack mounted, or portable applications in widely diverse, environmentally controlled locations (Class 5). The enclosure shall have a convertible/rack-mountable capability that have no openings on the top surface (Style E). The color of the enclosure shall be the color normally provided by the manufacturer (Color R). The type, class, style, and color requirements have been excerpted from MIL-T-28800 (Type II, Class 5, Style E, Color R). The equipments design and physical characteristics shall be such that controls, indicators, connectors, and switches are readily accessible by the user.

3.3.1 Materials. The contractor shall select the materials and parts not specified and the equipment shall meet all of the operational and environmental requirements specified herein. All replaceable assemblies, subassemblies, and parts of a designated model of a manufacturer shall be interchangeable on a noninterference basis with all such designated models provided by other

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manufacturers on previous contracts. Recovered/recycled materials shall be used to the maximum extent possible as long as they do not compromise the integrity of the equipment.

3.3.2 Components. The equipment shall not utilize externally accessible or externally removable assemblies/subassemblies (e.g. mainframe plug-in concept). All major components, assemblies, and subassemblies shall be identifiable by a part number placed in a discernible location.

3.3.3 Controls. All controls which are required to operate the equipment throughout its normal, specified performance characteristics, shall be located on the front panel. All controls, switches, and indicators shall be clearly marked to identify their function and operation. Unless otherwise specified in the contract or purchase order, built-in controls, adjustments and compensating devices used to calibrate, test, and maintain the equipment shall not be externally accessible.

3.3.4 Construction. The equipment shall be of solid-state, modular, miniaturized construction. The equipment shall be designed and constructed to comply with all requirements of performance, accuracy, and human engineering. The equipment shall provide adequate safety factors, meet the requirements of being easy to install and maintain, provide high reliability under varying operational environments, and conform to military service needs such as limited maintenance under continuous use for long periods of time. The equipment shall be constructed so that:

a. Subassemblies and chassis components can be removed without removing other hard wired subassemblies, printed circuit cards or components. Encapsulation and embedment (potting) of subassemblies shall not be used.

b. Calibration/adjustments can be made without removing components, printed circuit cards, or subassemblies. The use of extender cards is permitted for calibration, troubleshooting, or maintenance.

c. Printed circuit boards shall be designed to be removed without the need to unsolder cables and interconnecting wiring. Connections to all printed circuit cards shall be through pin and socket connectors. Printed circuit cards (mother boards) designed to distribute power and signals to other printed circuit cards (daughter boards) are excluded from this requirement. When mother boards of this type are used, they shall be accessible from both sides to allow for maintenance testing.

d. Indicator lights, other than light emitting diodes (LEDs), shall be accessible and replaceable from the operator's side of the front panel.

3.4 Electrical power sources and connections. The equipment shall operate from a nominal 115/230 volts, single phase, 50, 60 and 400Hz source. An input power selector with a provision to prevent accidental switching shall be provided.

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3.4.1 Leakage current. Leakage current shall not exceed 3.5 milliamperes (mA) if an open circuit should occur between an accessible part/component and ground. Accessible parts are defined as control shafts with knobs removed, switches, recessed controls, calibration and adjustment controls, and all other electro-mechanical parts/components which may pose a hazard to the operator/maintainer.

3.4.2 Fuses and circuit breakers. Fuses and circuit breakers shall be provided to disconnect the equipment from the power source in the event of excessive current drain. Either common or separate fuseholders may be provided. If only one fuseholder is used (common), the equipment shall be provided with the 115VAC fuse installed and the 230VAC fuse shall be stowed with the accessories.

3.4.3 Input power switch. A front panel mounted power switch shall be provided. Visual indication shall be provided on the front panel to show when the equipment is energized. Visual indication may be accomplished by digital displays, lamps, or other means which clearly indicate that the equipment is energized. The switch shall break both sides of the input power source.

3.4.4 Connectors. The input and output connectors shall be series "BNC" type female connectors except for the RF Output, which shall be a series "N" type female connector.

3.5 Enclosure requirements. The equipment shall be housed in an enclosure that has no openings on the top surface and provides protection from mechanical shock, and conditions associated with bench top use in environmentally controlled areas. The enclosure shall be capable of being rack mounted if specified in the contract or purchase order. The contractor shall supply a conversion kit to convert the enclosure for rack mounting. The conversion kit shall include all hardware required to configure the equipment for rack mounting. If the height or width of the front panel is less than the height or width of the rack, the conversion kit shall provide filler panels to accommodate the equipment. No part of the equipment shall protrude above or below the front panel when the enclosure is rack mounted.

3.6 Marking and identification. Reference designations shall be used to identify each part, subassembly, circuit board, and component. The marking/identification of these parts shall be placed so that they will be clearly visible without having to remove, unsolder, or disassemble the components/parts of the equipment. Warning markings shall warn of the location, nature, and extent of a hazard. Warning marking shall have a high contrast between the warning letters and the background colors, be located as close as possible to the hazard area, and be as permanent as the life expectancy of the equipment.

3.6.1 Identification plate. The equipment identification plate shall contain the following data:

a. Nomenclature

b. Manufacturer's name and code

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c. Equipment name and model number

d. Procurement instrument identification number (PIN)

e. Serial number

f. National stock number

g. Contract number

3.7 Environmental requirements. The equipment shall provide environmental resistance required for use in protected service areas. The enclosure shall be designed for bench top or rack mounted use with no openings on the top surface of the case. Panel mounted controls, indicators, switches, and fuseholders shall meet bench handling requirements without being damaged. Protection may be provided by the use of recessed front panel or rigid handle guards.

3.7.1 Electromagnetic interference. The equipment shall comply with the following emission and susceptibility requirements:

a. CE02, CE04 - Electromagnetic emissions in the frequency range of 10KHz to 50MHz shall not appear on ac or dc powerleads (see Fig. 1).

b. CS02 - The performance characteristics of equipment, systems, and subsystems shall not be degraded beyond the tolerances of the individual equipment specifications in the frequency range of 50KHz to 400MHz when subjected to the power levels specified in Figure 2.

c. CS06 - The test sample shall not exhibit any malfunction or degradation of performance beyond the tolerances given in the test sample's specification when the spike shown in Figure 3 is applied to the ac or dc power input lines.

d. RE02 - Narrowband E field emissions in the frequency range of 14KHz to 10GHz shall not be generated and radiated in excess of the values shown in Fig. 4. Broadband E field emissions (except from electric hand tools) in the frequency range of 14KHz to 10GHz shall not exceed the values of Figure 5.

e. RE02.1, RS03 - In the frequency range of 14KHz to 10GHz, the limits shall not exceed the values shown in Figure 6 and the tests shall be performed with RS03 at a susceptibility level of 1 volt per meter (v/m).

3.7.2 Humidity. The equipment shall meet the specified performance and accuracy requirements after being stored under the nonoperating conditions specified in Table 1. When the specified humidity is 95 percent ± 5 percent, the equipment shall withstand the effects of humidities up to 100 percent, including conditions where condensation occurs in and on the

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equipment. The equipment shall also conform to the performance and accuracy requirements under the operational temperature and humidity conditions specified in Table 1.

TABLE I. Temperature and humidity limits.

Temperature (°C)		RH for operating range (percent)
Nonoperating 1/	Operating	
-40 to 71	0 to 10	1/
	11 to 30	95 ±5
	31 to 40	75 ±5
	41 to 50	45 ±5

1/ Relative humidity not controlled

3.7.3 Vibration. The equipment shall conform to the specified vibration performance and accuracy requirements of Table II when tested in accordance with 4.5.4.

Table II. Vibration.

Vibration (5 to 55 Hz)				
	Time per Axis (minutes)			
Frequency (Hz)	Resonance Search	Cycling	Resonance Dwell	G Level at Maximum Frequency
5 through 55	15	15	10	2 at 55 Hz

3.8 Reliability requirements.

3.8.1 Reliability burn-in. Each signal generator delivered shall meet all requirements of this specification and shall be submitted to a 96 hour (minimum) on-time, burn-in procedure as specified in 4.5.6.1. The last 24 hours of burn-in shall be failure free.

3.8.2 Reliability. Reliability shall comply with requirements of this specification, contract, and/or purchase order. The specified MTBF shall be 2500 hours when tested as specified in 4.5.6. Any departure from the specified performance or operation shall constitute a failure and shall not be correctable by use of the operating controls after the test is initiated (see 6.4.2).

3.8.3 Maintainability The equipment shall be designed to permit isolation of faults, and repair down to the lowest discrete component (resistor, switch, transistor, etc) (see 6.4.3).

4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified are classified as follows:

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a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Mechanical and visual inspection. The equipment shall be subjected to a thorough mechanical and visual inspection to determine that all materials, workmanship, and safety characteristics comply with the specified requirements. Attention shall be given to the following specified requirements:

a. Completeness.

b. Nameplates, identification marking, labels.

c. Alignment and tightness.

d. Finish.

e. Fit and placement of components.

f. Operation of connectors, switches, controls.

g. Mountings and brackets.

h. Fastening and securing of devices and parts.

i. Accessibility of components and parts.

j. Welded and soldered joints and connections.

k. Forms, harnesses, and wiring.

l. Grounding connections.

m. Safety examinations.

n. Workmanship.

4.3 First article inspection. A first article inspection shall be performed by the contractor, after award of contract and prior to production. The first article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production.

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4.4 Conformance inspection. A description of the inspection procedure, sequence of inspections, number of units to be inspected, and the criteria for determining conformance to the qualification requirement shall be specified in the contract.

4.4.1 Inspection criteria. The first article inspection shall be performed on five units. The contractor shall prepare a test plan for both classes of inspection to include as a minimum the tests listed in TABLES III, IV, & V. Failure of any Group A through Group E test shall be counted as a major defect and may be cause for rejection of that equipment and could constitute rejection of the entire sample equipment submitted for first article inspection. The tests and inspections to be performed shall be identical for both classes of inspection.

4.4.2 Inspection sampling plan. The inspection sampling to be performed during both classes of inspection shall be in accordance with a contractor generated, Government approved sampling plan except as indicated below. Seals shall be used to control access to internal adjustments and circuitry on all reliability models. Breaking of the seals to make adjustments (or repair) shall be observed as a failure and the elapsed failure-free operational period with unbroken seals shall be observed to establish the calibration interval.

4.4.2.1 First production lot. The Group C, D, E & F conformance inspections shall be performed on the first production lot only.

TABLE III. Examination and test groups.

Description	Requirement	Test method
Group A		
Preoperational inspection	3.3.4	4.2
Leakage current	3.4.1	4.2
Level A performance	Table IV	4.2
Group B		
Level B performance	Table V	4.2
Group C		
Electrical power	3.4	4.2
Environmental requirements	3.7	4.2
Humidity	3.7.2	4.2
Vibration	3.7.3	4.2
Group D		
Electromagnetic interference	3.7.1	4.4.2.2
Group E		
Dimensions	3.1.1.1	
Weight	3.1.1.1	4.2
Front panel marking	3.6	4.2
Group F		
Reliability	3.8	4.2, 4.4.2.2

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TABLE IV. Level A performance tests.

Description	Requirements
Output frequency	3.1.1
Amplitude modulation	3.1.4
Amplitude modulation - internal	3.1.4
Pulse modulation (PM)	3.1.6
FM deviation	3.1.7
RF voltage output	3.1.8
Output attenuation	3.1.9
Output impedance	3.1.10
Input impedance	3.1.11

NOTE: Level A testing is abbreviated testing (see 6.4.1)

TABLE V. Level B performance tests.

Description	Requirements
Output frequency accuracy	3.1.3
Output frequency response	3.1.3.1
RF spurious output	3.1.3.2
RF noise level	3.1.3.3
External AM sensitivity	3.1.4.1
Percent modulation	3.1.4.2
Modulation distortion	3.1.5.1
FM internal spurious AM	3.1.7.1
Carrier frequency shift	3.1.7.2
Modulation voltage	3.1.7.3
Standing wave ratio (SWR)	3.1.8.1

4.4.2.2 Group D and F tests. Group D electromagnetic interference (EMI) tests and the Group F Reliability test shall be performed during quality conformance inspection only if not performed during first article or bid sampling testing.

4.5 Conformance tests.

4.5.1 Open circuit voltage test (see Figure 7 and 8). The open circuit voltage test shall be measured on equipment in the normal operating configuration. Equipment controls in each operation mode shall be such that maximum power will be utilized during the open circuit voltage

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test. A true rms voltmeter shall be used to measure ac potentials, and a dc voltmeter with a 10 megohm or greater input impedance shall be used to measure dc potentials. The probe in Figures 7 and 8 shall be used to measure voltage on any conducting surface of the equipment. Measurement shall be made from controls and switches with knobs removed. Voltage shall be measured from each part to all other simultaneously accessible parts for every combination of switch positions specified in Figures 7 and 8. The open safety ground connector shall be reconnected immediately after the tests have been completed.

4.5.2 Leakage current test (see figures 9 and 10). Each equipment connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface with all safety and ground conductors between the equipment intact. The safety ground conductor between the equipment and the power source shall be opened during the test. The equipment shall be connected as shown in Figure 9 if it is connected to a single phase ac power source, or as shown in Figure 10 if the equipment is connected to a dc power source.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.2.1 Measurement. The leakage current test shall be performed on equipment in the normal operating configuration. Equipment controls in each operation mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method, using a true rms voltmeter for ac potentials and a high impedance dc voltmeter for dc potentials. The voltage measured across a 1500 ohm resistor (see Figures 7 and 8) shall not exceed 5v at the highest nominal power line voltage and the highest and lowest nominal power line frequencies.

4.5.3 Electromagnetic interference. One equipment shall be subjected to an EMI test for compliance with the requirements of 3.7.1. All test set-ups and procedures shall be performed in accordance with a contractor prepared and Government approved EMI test plan. MIL-STD-462 can provide guidance but its use is not mandatory. All emission and susceptibility tests shall be performed with the supplied output cable connected to, and extended parallel with, the front of the equipment. The output cable shall be terminated to simulate a normal loading configuration.

4.5.3.1 Radiated emissions testing. The equipment shall pass the radiated emissions test only if the emitted radiation is below the limits specified in 3.7.1.

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4.5.3.2 Radiated susceptibility testing. In performing this test the equipment shall be oriented in four different positions. These positions shall be:

a. Horizontal 0° - The front of the equipment shall face the source of radiation.

b. Horizontal 90° - The right side of the equipment shall face the source of radiation (the equipment is rotated 90° clockwise around a vertical axis).

c. Horizontal 180° - The rear of the equipment is facing the source of radiation.

d. Horizontal 270° - The left side of the equipment is facing the source of radiation (the equipment is rotated 270° clockwise around a vertical axis).

4.5.4 Humidity. The equipment shall be tested in accordance with contractor developed Government approved test plans and shall meet the specified requirements of 3.7.2. MIL-T-28800 can be used as guidance for humidity testing but its use is not mandatory.

4.5.5 Vibration. The equipment shall be tested in accordance with contractor developed, Government approved test plans and shall meet the requirements of 3.7.3. MIL-T-28800 (for Class 5 equipment), can be used as guidance for the vibration testing but its use is not mandatory. The satisfactory operation test shall be performed prior to and following vibration testing.

4.5.5.1 Test failures. If an equipment failure should occur, the equipment under test may be repaired at the discretion of the procuring activity. If repair is allowed, testing will continue from the point at which the failure occurred until the remaining test period is completed. The portion of the test prior to the failure will be repeated to evaluate the integrity of the repair. Failure modes that are not related to the original failure will be disregarded during the retest.

4.5.6 Reliability. The reliability tests shall be performed as follows:

4.5.6.1 Burn-in. Each deliverable equipment shall be subjected to a minimum 96 hours on-time burn-in period, prior to Group A testing. During the last 24 hours of burn-in, the equipment must operate failure free. Up until this time, equipment will be allowed to accumulate failures. Each equipment which fails during the final 24 hour period shall be repaired and returned to test until it successfully survives a 24 hour period without failure. Failures which occur during the burn-in test shall be noted and reported, but shall not count toward the establishment of equipment MTBF. Prior to burn-in, the satisfactory operation test of 4.2 shall be conducted. Daily satisfactory operation checks shall be conducted. For the last 24 hour failure free period, a complete satisfactory operation test shall be conducted prior to, and after the period.

4.5.6.2 Reliability sampling plan.

4.5.6.2.1 First article inspection samples. The Group F reliability demonstration shall be conducted on at least five (5) samples. The use of ten (10) samples is encouraged, but not more than ten (10) samples shall be used.

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4.5.6.2.2 Conformance inspection samples. The Group F reliability tests shall be performed on the first production lot, only if not performed during First Article testing. From those units of the first lot that have passed the required Group A and B tests, a random sample of ten (10) units shall be selected, for Group F testing.

4.5.6.2.3 Reliability test plan. The reliability tests shall be conducted in accordance with the following test plan:

TABLE VI. Reliability test plan.

Number of Failures	Accept** (Equal or More)	Total Test Time* Reject*** (Equal or Less)
0	2000	N/A
1	3750	558
2	5375	2063
3	6875	3850
4	8375	5725
5	9875	7725
6	11,375	11,375 ****

* Total test time is total unit hours of "equipment ON" time (in hours).
** Accept if test time is greater than or equal to that listed.
*** Reject if test time is less than or equal to that listed.
**** Reject if test time is strictly less than 11,375 with 6 failures.

4.5.6.3 Test length. Testing shall continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with the specified test plan. Only equipment "ON" time may be used in MTBF or longevity determinations. Testing shall be monitored in such a manner that the times to failure may be estimated with reasonable accuracy. No single equipment "ON" time shall be less than one-half the average operating time of all equipments "ON" test.

4.5.6.4 Test conditions.

4.5.6.4.1 Operating mode (duty cycle). Prior to the burn-in, a satisfactory operation check shall be conducted. For the last 24 hour failure free period, a complete satisfactory operation test shall be conducted prior to and after the period.

4.5.6.4.2 Test level. While under test, the equipment shall be subjected to the conditions specified in section 6.5.

4.5.6.4.3 Input voltage cycling. When so directed by the procuring activity, voltage cycling shall be accomplished as follows: The input voltage shall be maintained at one hundred ten

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percent (110%) nominal voltage for one-third of the equipment "ON" cycle, at the nominal value for the second one-third of the equipment "ON" cycle, and at ninety percent (90%) for the final one-third of the equipment "ON" cycle. This cycling procedure is to be repeated continuously throughout the reliability test.

4.5.6.5 Reliability examination and test method. The following inspections shall be used to verify equipment operation during reliability testing. The satisfactory operation test of 4.2 shall be used to confirm proper equipment operation prior to and following reliability testing. The satisfactory operation test shall also be performed weekly to verify equipment operation. The satisfactory operation check of 4.2 shall be used to monitor proper operation of the equipment daily.

4.5.6.6 Reliability failure actions, group F. In the event of failure(s) during reliability testing, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the testing process and repaired with a minimum interruption of the equipments that are continuing in the testing cycle. All failed parts shall be replaced and after the failed equipment has been repaired and is operable, it shall be returned to the testing cycle with appropriate entries made in the data log.

4.5.6.6.1 Failure categories. The failure categories and required actions are:

a. Relevant failures. All failures are relevant unless it is determined by the procuring activity to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service. Failures chargeable to workmanship deficiencies in the equipment shall be counted as relevant.

b. Multiple failures. In the event of simultaneous part failures, each failed part which will independently prevent satisfactory equipment performance shall be counted as an equipment failure. All relevant failures shall be counted as equipment failures and included in the MTBF calculations. At least one equipment failure shall be counted when a dependent failure is claimed.

c. Pattern failures. When a definite pattern of independent or dependent failures is identified, the equipments shall be subjected to corrective action in accordance with 4.5.6.6.

4.5.6.6.2 Analysis of failures. The cause of each equipment or part failure, including Government furnished equipment (GFE) or parts to be included in or as a part of the test group equipment, shall be determined by investigation and analysis. The investigation and analysis shall consist of any applicable method (such as test, application study, dissection, X-ray analysis, microscopic analysis, or spectroscopic analysis) which may be necessary to determine the cause of failure. No part or unit shall be replaced in any equipment undergoing reliability testing unless the failed part or unit can be unmistakenly demonstrated by the repair activity to be outside of the specification tolerances. Details in the investigation and analysis shall be referred to the design or quality control activity for further study and analysis.

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4.5.6.6.3 Failure confirmation. All failures observed during the equipment reliability testing shall be confirmed. Failure to do so is insufficient grounds for deletion of such failures in the count of total failures. Lack of failure confirmation shall instigate a close review of the test facility. If the test facility can be shown to be both at fault and completely accountable for the failure to the satisfaction of the procuring activity, then and only then can the count of such a failure be eliminated.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended use. Signal generator SG-1170/U is a rugged, solid state device that serves as an extremely accurate frequency reference to provide highly stable signals over its specified frequency range. The equipment is intended for use in maintaining electronic communications equipment by equipment repair units in intermediate and depot maintenance.

6.1.1 Equipment replaced. Equipment procured according to this specification is intended to replace all versions of various older equipments now fielded. Equipments to be replaced are:

a. AN/RUM-70

b. AN/USM-44

c. AN/USM-103

d. AN/USM-252

e. SG-12

f. SG-1038

g. TS-497

h. SG-309

i. SG-975

6.2 Acquisition requirement. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Packaging requirements (see 5.1).

c. When rough handling and functional tests are required.

d. Place of final inspection.

e. Technical literature required.

f. Quantity of tools and running spare parts required.

g. Marking for shipment and shipping containers.

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h. Test plans and test reports.

i. Classification of inspection and number of samples required.

j. Rack mounting requirements.

k. Alternate power requirements.

1. Maintainability rationale.

m. Reliability rationale.

n. Non use of radio active materials.

6.3 Contract data requirements.

a. Nomenclature assignment

b. National stock number

c. Equipment sample test plans

d. Pretest performance records

e. First production test data

f. Identification plate drawing

g. Maintenance Allocation chart (MAC)

6.4 Definitions.

6.4.1 Level A performance tests. Level A testing (TABLE IV) is a reduced amount of testing which is performed on each equipment produced. Its purpose is to insure that all functions and modes of operation of the equipment are evaluated without extensively checking each parameter as required in the Level B test (TABLE V). The approved equipment test procedure should specify the actual amount of testing to be performed.

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6.4.2 Reliability rationale. Reliability rationale submitted with the bid should provide clear and concise rationale showing how the reliability of the equipment complies with minimum requirements of the solicitation. This may include data from previous reliability tests, reliability predictions and other data available to the offeror. In the absence of such data on the equipment being offered, such data on similar equipment of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accomplished with specifications and technical literature on the similar equipment. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production reliability testing.

6.4.3 Maintainability rationale. Maintainability rationale submitted with the bid should provide clear and concise rationale showing how the maintainability of the equipment complies with requirements of the solicitation. This may include data from previous maintainability tests, records of repair and calibration data, and other data on the equipment available to the offeror. Data on selection of module size and other design features relevant to maintainability may also be submitted.

6.5 Reliability production acceptance test (excerpted from MIL-STD-781). The reliability production acceptance tests should demonstrate the adequacy of the mean-time-between-failure (MTBF) of the equipment either initially, after a production or design change, or after a reject decision. The minimum consecutive quantity of equipments required to meet qualification is determined by the total quantity of equipment on contract or purchase order (see TABLE VII).

TABLE VII. Qualification equipment quantity.

	Minimum Consecutive Quantity Required		
Quantity	A	B	C
1-8	All	N/A	N/A
9-10	8	Balance of lot	Balance of lot
11-18	9	Balance	Balance
14-17	10	Balance	Balance
18-23	11	18 or balance	Balance
24-30	12	21 or balance	28 or balance
31-35	13	22 or balance	30 or balance
36-42	13	28	32 or balance
43-50	14	24	34 or balance
51-60	14	25	35 or balance
61-80	14	26	36 or balance
81-100	15	27	37
101-200	15	28	40
Over 200	15	29	42

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a. Column A presents the quantity required when there has been no reject decision for the prior test group. Column A is used for the initial qualification test or after a production or engineering change.

b. Column B presents the quantity required when, based on all of the test data, there has been a reject decision for the prior test group and one equipment therein has been identified by data censorship techniques to be unsatisfactory.

c. Column C presents the quantity required when, based on all of the test data:

(1) There has been a reject decision for the prior test group and two or more equipments therein have been identified by data censorship techniques to be unsatisfactory.

(2) The two prior test groups each have experienced a reject decision regardless of the number of unsatisfactory equipments identified therein.

6.5.1 Sampling phase of reliability production acceptance tests. The sampling phase, as distinguished from the qualification phase, is a periodic series of tests to indicate continuing production of equipment meeting the specified MTBF requirement. In general, since the sampling phase requires test of fewer equipments, the low consumer's risk of the qualification phase is not maintained. The right to apply the sampling phase is earned only by passing the qualification phase.

6.5.2 Quantity required for the sampling phase. The quantity required for the sampling phase should be determined by the contractor and should lie in a range defined as follows:

a. The upper limits should be determined by the specified MTBF and an adjusted production rate. The adjusted production rate is determined by multiplying the expected production rate by a factor of 500 divided by the average number of operating hours of test per equipment per month.

b. The lower limit should be determined by the specified MTBF and the average number of operating hours of test per equipment per month.

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6.6 Subject (keyword) listing.

Amplitude modulation (AM)
Carrier frequency shift
Frequency modulation (FM) deviation
Modulation distortion
Percent modulation
Pulse modulation
RF spurious output
Standing wave ratio (SWR)

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - CR

Preparing activity
Army - CR

(Project 6625-A117)

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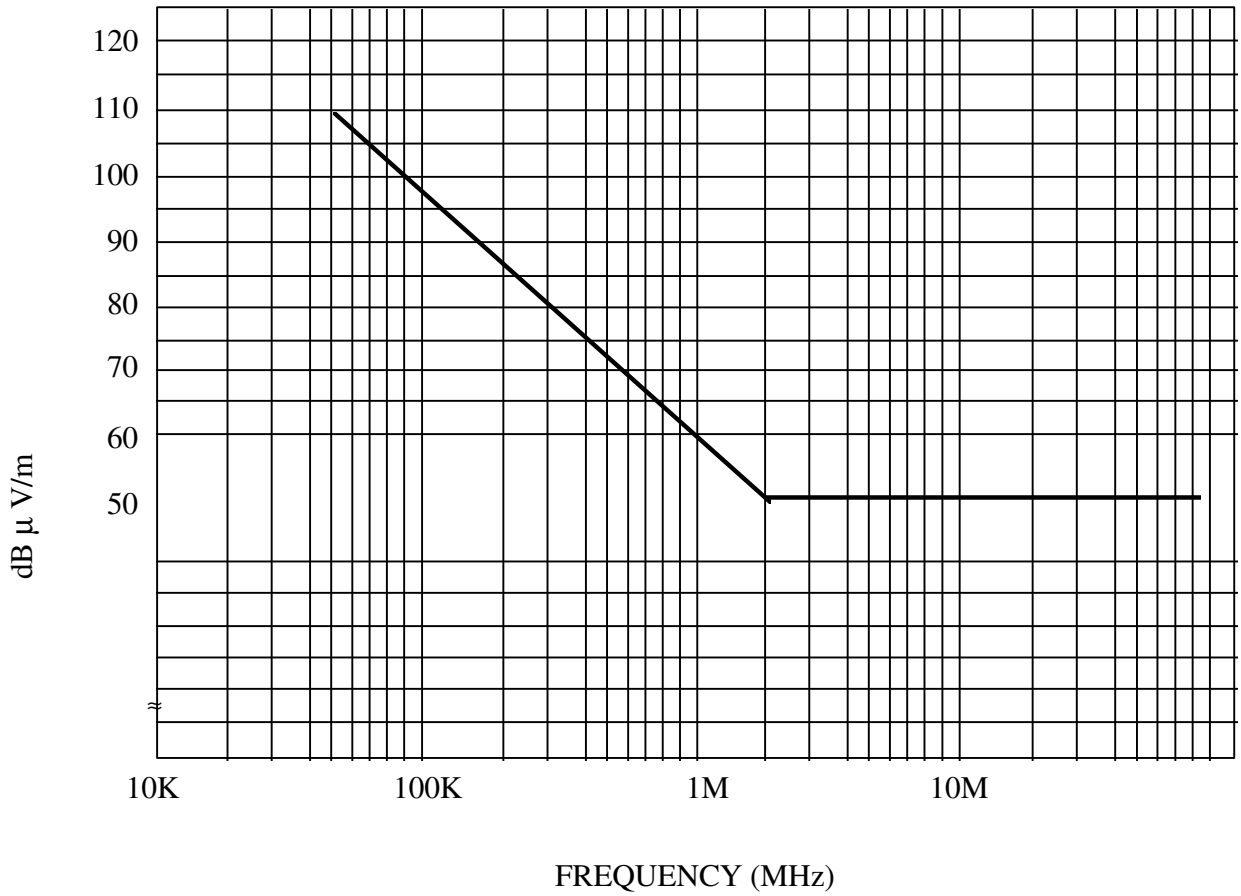


FIGURE 1. Limit for CE02 & CE04 broadband emissions.

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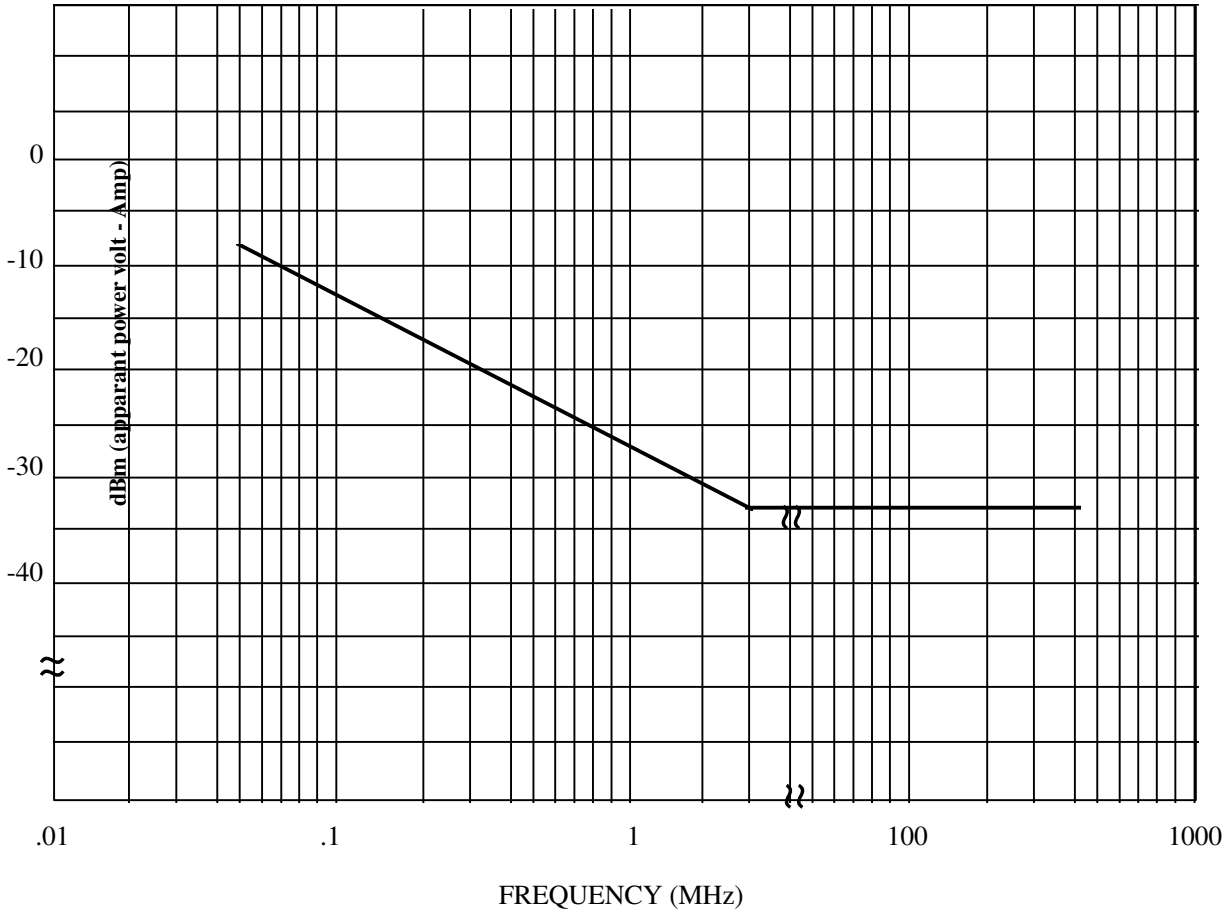
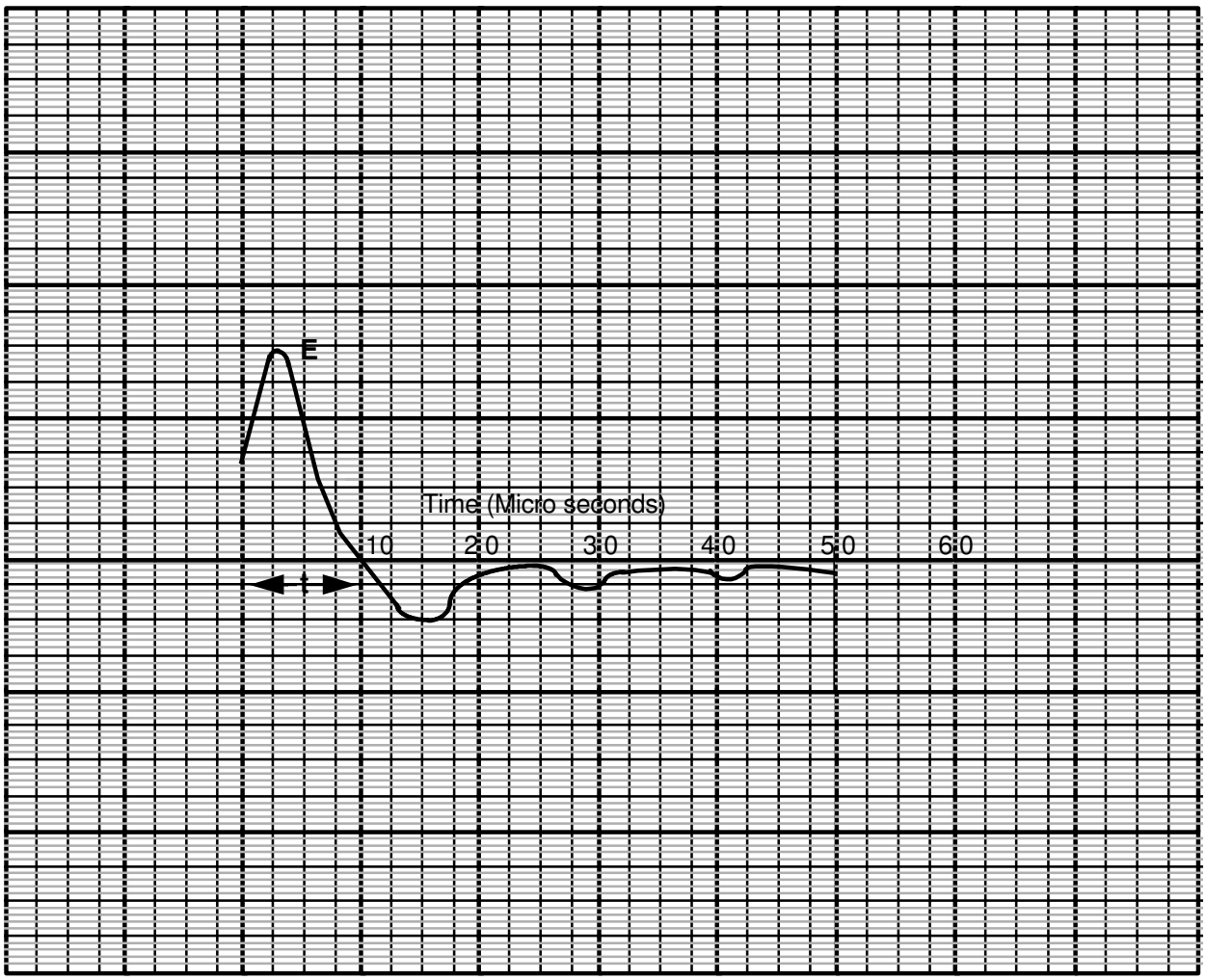


FIGURE 2. Limits for CS02.

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E = 2 TIMES LINE VOLTAGE OR 100 VOLTS, WHICHEVER IS LESS
t = 10 MICROSECONDS

FIGURE 3. Limits for CE06.

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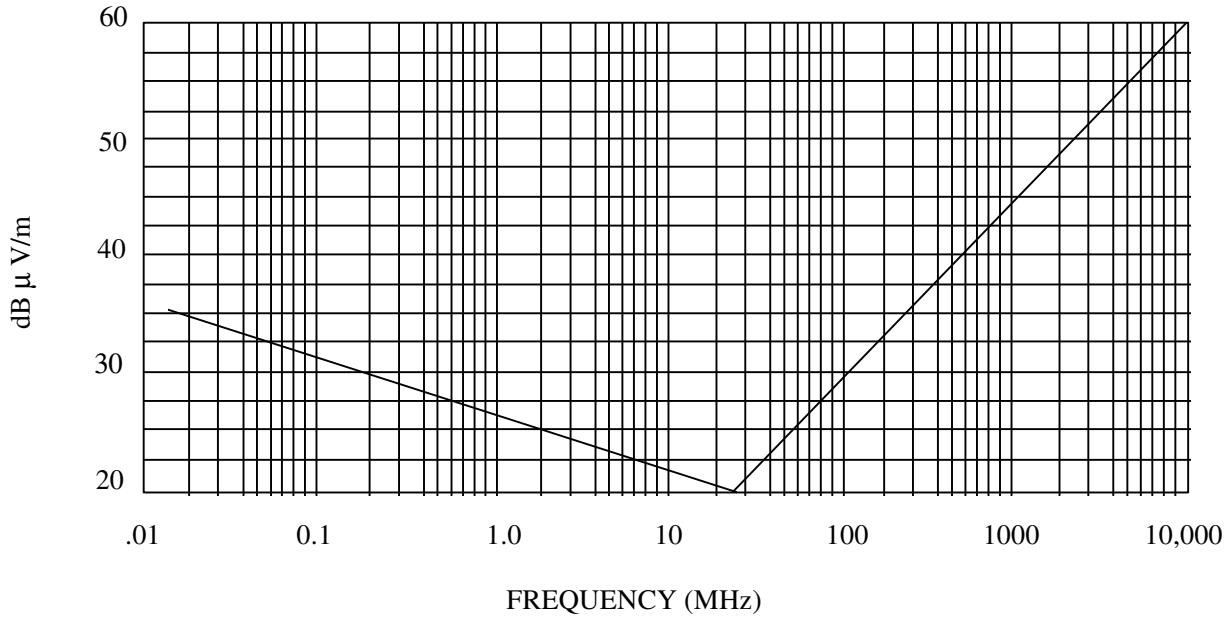


FIGURE 4. Limit for RE02 narrowband emission.

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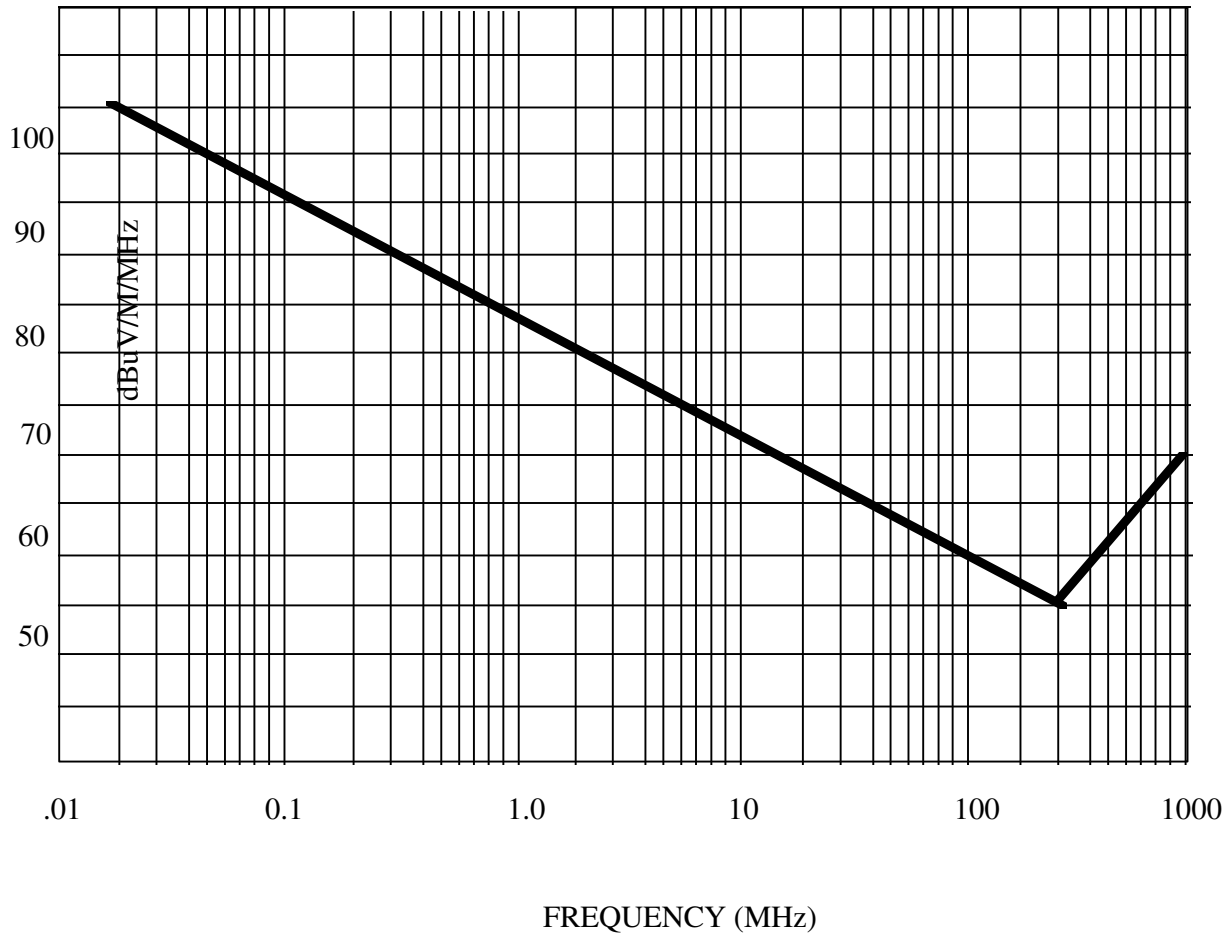


FIGURE 5. Limit for RE02 and RE05 broadband emissions.

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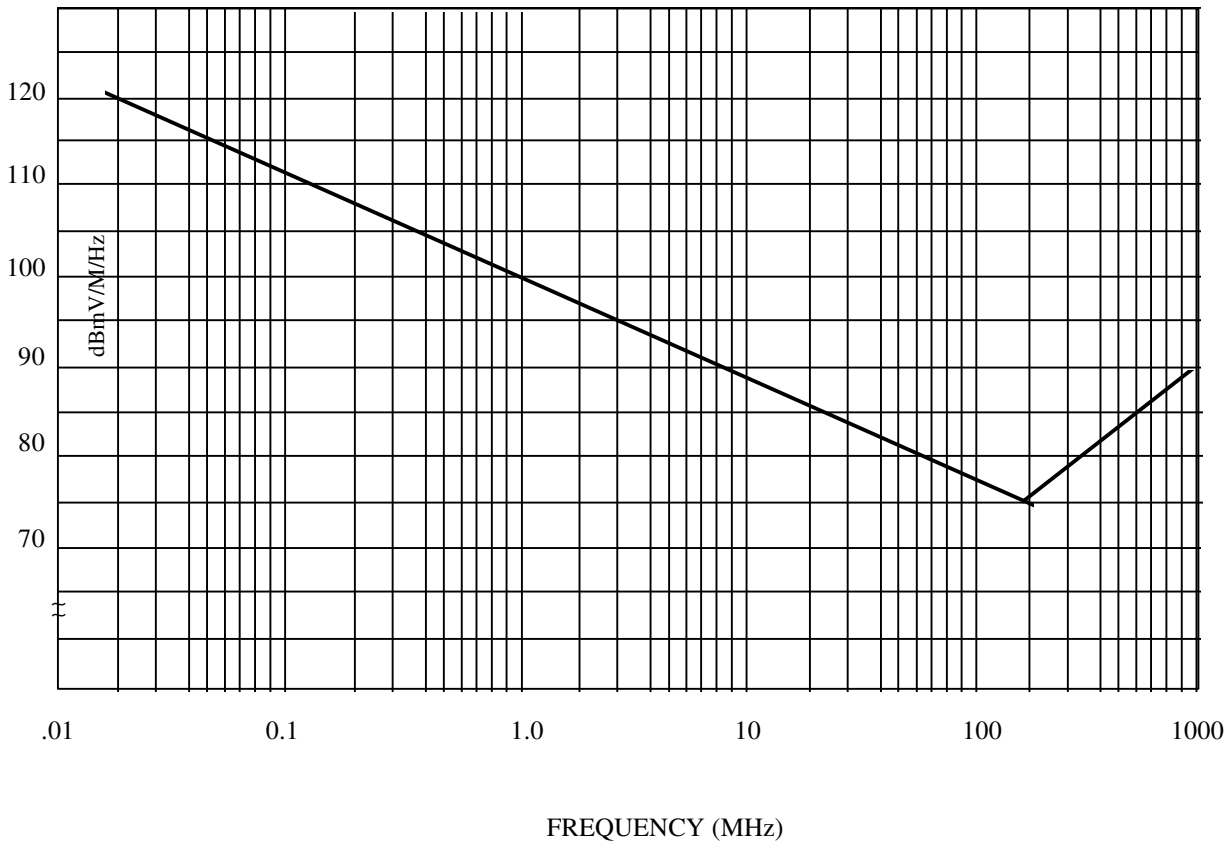


FIGURE 6. Limit for radiated switching transient.

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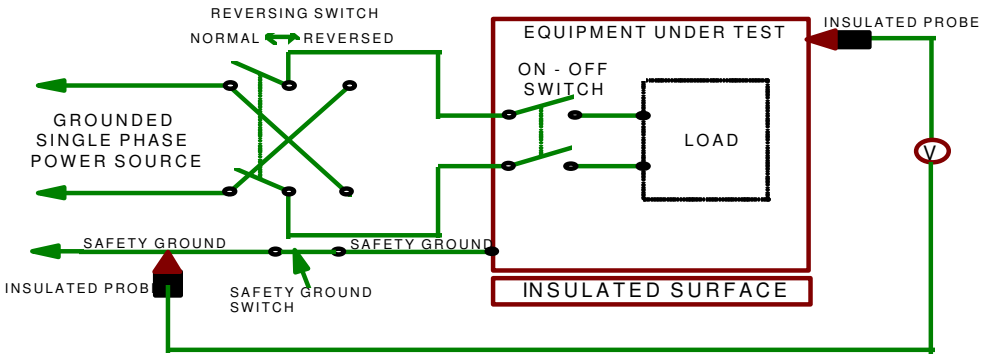


FIGURE 7. Single phase AC, open circuit voltage test schematic.

WARNING DO NOT TOUCH EXPOSED METAL SURFACES THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR SINGLE PHASE LEAKAGE CURRENT MEASUREMENT.

General Order of Test:

1. Power source off, connect the equipment as shown in Figure 7.

2. On-off switch-off, safety ground switch-closed. Reversing switch normal. Connect power source.

3. **OBSERVE WARNING!** Safety ground switch-open, on-off switch-on.

4. For each probe point, record voltmeter reading (case, connectors, controls, shafts, etc).

5. On-off switch-off. Repeat step 4.

6. Reversing switch to the reversed position. On-off switch-on. Repeat step 4.

7. On-off switch-off. Repeat step 4.

8. Safety ground switch closed. Reversing switch in the normal position.

9. Repeat steps 3 through 8 for each mode of operation.

10. Remove power source. Disconnect equipment.

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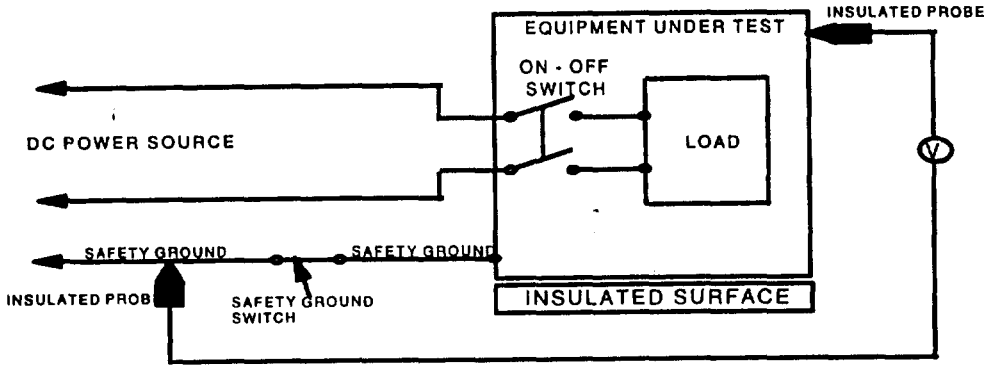


FIGURE 8. DC power source, open circuit voltage test schematic.

WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR SINGLE PHASE LEAKAGE CURRENT MEASUREMENT.

General Order of Test:

1. Power source off, connect the equipment as shown in Figure 8.
2. On-off switch-off, safety ground switch-closed. Connect power source.
3. **OBSERVE WARNING!** Safety ground switch-open, on-off switch-on.
4. For each probe point, record voltmeter reading (case, connectors, controls, shafts, etc).
5. On-off switch-off. Repeat step 4.
6. Safety ground switch on. On-off switch-on. Repeat steps 4 and 5.
7. Remove power source. Disconnect equipment.

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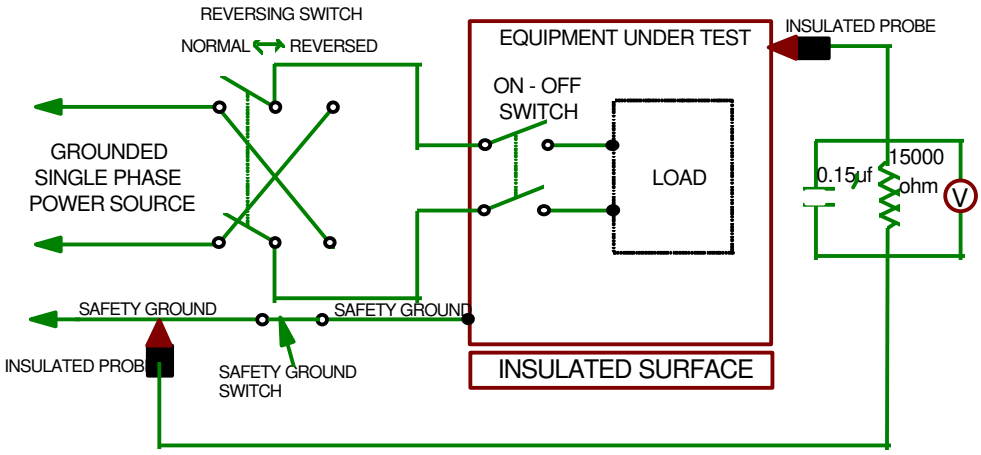


FIGURE 9. Single phase AC, leakage current test schematic.

WARNING DO NOT TOUCH EXPOSED METAL SURFACES
THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR SINGLE PHASE LEAKAGE CURRENT MEASUREMENT.

General Order of Test:

1. Power source off, connect the equipment as shown in Figure 9.

2. On-off switch-off, safety ground switch-closed. Reversing switch normal. Connect power source.

3. **OBSERVE WARNING!** Safety ground switch-open, on-off switch-on.

4. For each probe point, record voltmeter reading (case, connectors, controls, shafts, etc).

5. On-off switch-off. Repeat step 4.

6. Reversing switch to the reversed position. On-off switch-on. Repeat step 4.

7. On-off switch-off. Repeat step 4.

8. Safety ground switch closed. Reversing switch in the normal position.

9. Repeat steps 3 through 8 for each mode of operation.

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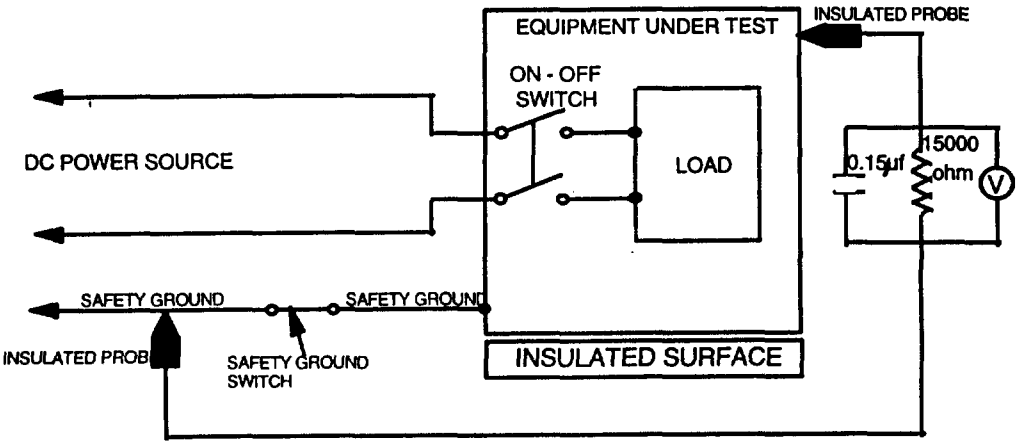


FIGURE 10. DC power source, leakage current test schematic.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR SINGLE PHASE LEAKAGE CURRENT MEASUREMENT.

General Order of Test:

1. Power source off, connect the equipment as shown in Figure 8.
2. On-off switch-off, safety ground switch-closed. Connect power source.
3. **OBSERVE WARNING!** Safety ground switch-open, on-off switch-on.
4. For each probe point, record voltmeter reading (case, connectors, controls, shafts, etc).
5. On-off switch-off. Repeat step 4.
6. Safety ground switch on. On-off switch-on. Repeat steps 4 and 5.
7. Remove power source. Disconnect equipment.

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3. DOCUMENT TITLE GENERATOR, SIGNAL, VHF, SG-1170/U

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
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